

Interactive Wall

Overview

This project presents a **low-cost interactive wall system** that allows users to control presentation slides using **hand gestures** instead of a mouse or keyboard.

The system uses a webcam to track the user's hand and maps finger movements to screen regions, enabling:

- Next slide
- Previous slide
- Exit program

It is designed to be **real-time, user-friendly, and accessible**, using only standard hardware and open-source libraries.

Features

- 🖐️ Hand gesture-based slide control
- 📷 Real-time hand tracking using MediaPipe
- 🎯 Region-based interaction (simple & efficient)
- ⚡ Fast response time (~115 ms)
- 💻 Works with standard webcam (no special hardware needed)
- 🔄 Smooth and stable tracking using filtering algorithm

Technologies Used

- Python 3.9+
- OpenCV
- MediaPipe
- NumPy
- PyAutoGUI

System Requirements

- OS: Windows 10 or higher
- Webcam

- Projector or display screen
- Python 3.9+ installed

Installation

1. Clone the repository:

```
git clone https://github.com/maan-magid/Interactive\_Wall  
cd interactive-wall
```

2. Install required libraries:

```
pip install opencv-python mediapipe numpy pyautogui
```

How to Run

1. Open terminal / command prompt
2. Navigate to project folder
3. Run the program:

```
python Run.py
```

Camera Setup (IMPORTANT)

For best performance:

- Place camera **1 to 1.5 meters** from the wall
- Ensure camera captures **only the projection area**
- Adjust field of view (recommended $\sim 65^\circ$)
- Make sure lighting is **good and stable**

 The 4 corners of the camera view should match the screen.

How to Use

After running the program:

1. Calibration

- Follow on-screen instructions
- Point your finger to the **4 corners**
- Hold briefly to register each point

2. Interaction Mode

Move your index finger to control slides:

-  Right side → **Next slide**
-  Left side → **Previous slide**
-  Top-left corner → **Exit system**

Control Behavior

- The system includes a **delay (~3 seconds)** between slide changes
- This prevents accidental multiple triggers

Performance

- Accuracy: **~94%**
- Response Time: **~115 ms**
- Frame Rate: **20–30 FPS**

Limitations

- Sensitive to poor lighting
- Fast hand movement may reduce accuracy
- Requires calibration before use

Tips for Best Performance

- Use a **plain background**

- Keep hand movements **smooth**
- Ensure proper calibration
- Avoid shadows or clutter

Troubleshooting

Problem	Solution
Hand not detected	improve lighting / move closer
Slides skip too fast	Wait for delay between gestures
Program not working	Check Python & libraries installed
Camera not detected	Ensure no other app is using it

Future Improvements

- Multi-gesture support (zoom, draw, etc.)
- Multi-user interaction
- Better performance in low light
- AI-based gesture recognition

Author

Developed by: **Maan Mohamed**